

LC 2521: Distinct Prime Factors of Product of Array

Concept & Template

To find prime factors of any number remember this template:

let the no. for which we need prime factors be x then

```
int temp = x;
for(int p=2; p*p <= temp; p++){
    if (temp % p == 0) {
        primes.push_back(p);
        // now remove copies
        while (temp % p == 0) {
            temp /= p;
        }
    }
}
```

now if any number remains, it itself is a prime, so

```
if(temp > 1) primes.push_back(temp);
```

Handling Large Products

Now the product can exceed the range so we won't simply find the product and see its unique prime factors.

Complete Implementation

```
class Solution {
public:
    int distinctPrimeFactors(vector<int>& nums) {

        set<int> primes;
        for(int x:nums){

            int temp=x;
            for(long long p=2;p*p<=temp;p++){
                if(temp%p==0){
                    primes.insert(p);
                    while(temp%p==0){
                        temp/=p;
                    }
                }
            }

            if(temp>1){
                primes.insert(temp);
            }
        }

        return primes.size();
    }
};
```